

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output is intent class (closed 40-class label) plus structured slot-tag spans in \$\$-delimited text [Q145, Q146], scored with accuracy and F1 [Q84]. This matches the deployment's task-oriented dialogue NLU output requirements directly. Best fine-tuned models reach 93.7% intent accuracy / 85.6% F1 [Q19, Q78], setting an upper bound; Amharic-specific scores are reported in Table 5 (per-language) but not extracted in registry — gap noted. One minor concern: the eng config used for cross-lingual transfer contains lossy transliterations of Ethiopian entities (e.g., 'Dashen' → 'dasion') [DATASET-D34], which slightly degrades cross-lingual transfer evaluation quality. OF was marked LOWER priority and substantively the form fits.

ID	Evidence
Q19	"Our result shows that fine-tuning baselines could reach an accuracy of 93.7% and F1-score of 85.6 for intent detection and slot-filling tasks respectively." (p.2)
Q78	"Overall, AfroXLMR-76L achieves the best performance on both ID-SF tasks, with an average accuracy of 93.7% and an F1 score of 85.6%, respectively." (p.6)
Q84	"Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages." (p.7)
Q145	"Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return '\$\$' only." (p.19)
Q146	"Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater" (p.19)
D34	raw: "እባክህ ደግሞ ለዚህ ባንክ ልክህኝ?" / text: "can you please check my dasion bank account" — "Dashen Bank" (major Ethiopian bank) — Amharic source is culturally grounded but English transliteration is lossy ("dasion")

Strengths

- Structured intent + slot-tag output format directly matches deployment NLU module requirements [Q145, Q146]
- Accuracy and F1 metrics appropriate for closed-class intent and span-based slot evaluation [Q84]
- Amharic post-review annotation quality high ($\kappa=0.933$) [WEB-17]

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WEB-17	Amharic post-review $\kappa=0.933$ supports output reliability (https://www.researchgate.net/publication/389056008_INJONGO_A_Multicultural_Intent_Detection_and_Slot-filling_Dataset_for_16_African_Languages)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — text-based intent class + slot-tag output exactly matches deployment NLU module needs [Q145, Q146].

OF-2: TTS not relevant — deployment is text-based smartphone app.

OF-3: Smartphone-app users are positively selected for literacy [WEB-4 ~15% smartphone ownership]; text output is appropriate. National literacy ~71% [WEB-1, WEB-2] is lower than the smartphone-using subset, but for the deployment's user segment text output is suitable.

OF-4: Minor: lossy transliteration of Ethiopian entities in the eng config [DATASET-D34] could weaken cross-lingual transfer evaluation but does not affect direct Amharic deployment evaluation.

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Q84	"Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages." (p.7)
Q145	"Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return `` only." (p.19)
Q146	"Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater" (p.19)
D34	raw: "እባክህ ያዳኸን ባንክ ሂሳቤን ታረጋግጡልኛለችሁ?" / text: "can you please check my dasion bank account" — "Dashen Bank" (major Ethiopian bank) — Amharic source is culturally grounded but English transliteration is lossy ("dasion")
WEB-1	https://statbase.org/data/wld-literacy-rate/
WEB-2	https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=ET
WEB-4	Ethiopia smartphone ownership ~15% — text-based output appropriate for app user segment (https://birmetrics.com/ethiopia-narrows-mobile-gender-gap-to-24-but-smartphone-access-for-women-remains-just-6/)

Information Gaps

- Amharic-specific intent accuracy and slot F1 from Table 5 of the paper not directly extracted

Remediation

Gap 1: Lossy transliteration of Ethiopian entities in the eng config used for cross-lingual transfer (e.g., Dashen Bank → 'dasion')

Recommendation: If using INJONGO's English split as a cross-lingual source, normalize transliterations of Ethiopian named entities to standard English spellings (Dashen Bank, Awash Bank, Abay Bank) before training; otherwise prefer in-language SFT.